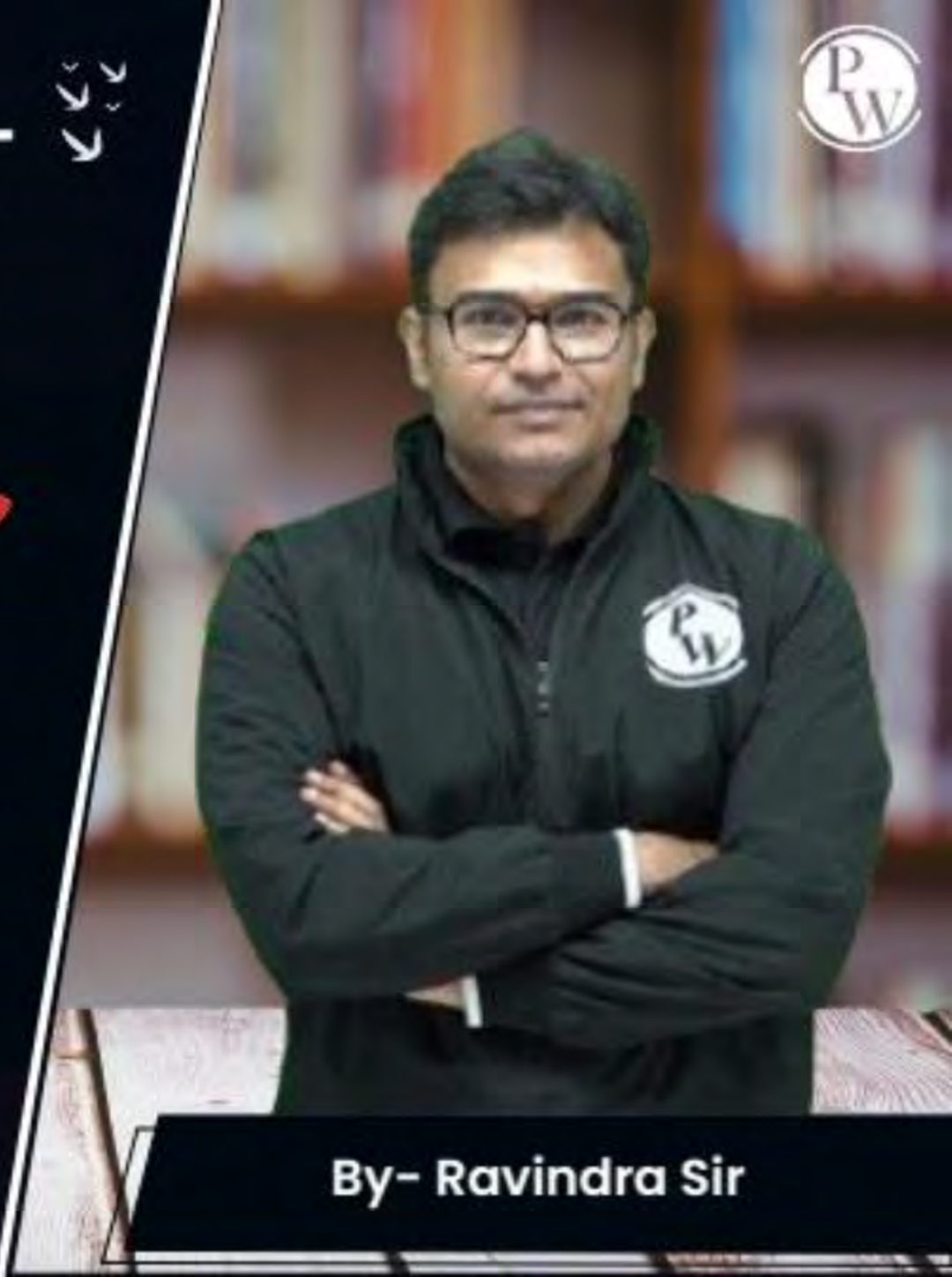
Computer Science & IT

Theory of Computation



Regular languages

Lecture No. 08



By- Ravindra Sir

Recap of Previous Lecture



Topic DFA

Topic NFA

Topic RE

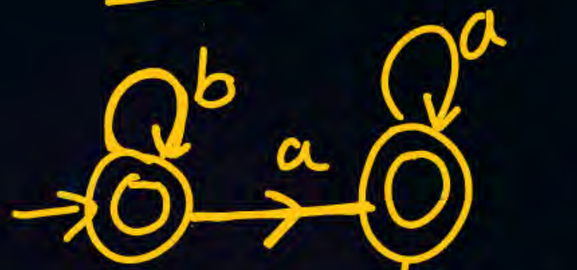
Topics to be Covered



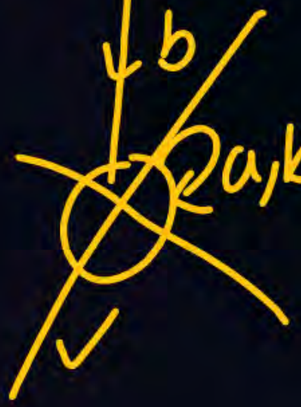
- Topic — DFA
- Topic — NFA
- Topic — RE

a is never followed by a 'b'

b^*a^*





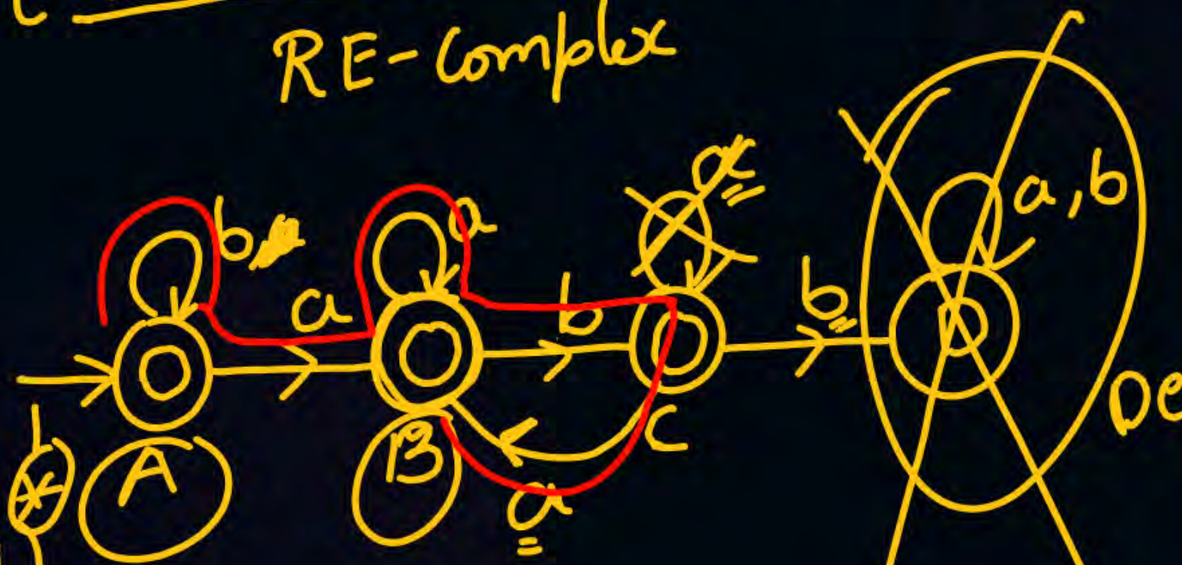


{ϵ, a, aa, aaa ....
ϵ, b, bb, bbb ....
ba, bba, baa ....}

If a in plant a should never be followed by bb

$L = \{\epsilon, b, bb, bbb \ldots\ldots; a, aa, aaa \sim\sim, ab, bab, aba \ldots\ldots\}$

RE - complex



b^*a^*b

abab

ab(bbb)

abbbb

abbb

X

$\beta \rightarrow$ check $a \rightarrow bb$

$(a+b)^*$

abbbb

Standard languages :-

$(a+b)^*$

$a^* b^*$

$a^+ b^+$

$a^* b^+$

$a^+ b^+$

$a^* + b^*$

$a^n b^m / n, m \geq 1$

$a^+ b^+$

$aa^* bb^*$ starting with 'a'

$L = \{ ab, aab, abb, aaab \ldots\ldots \} =$

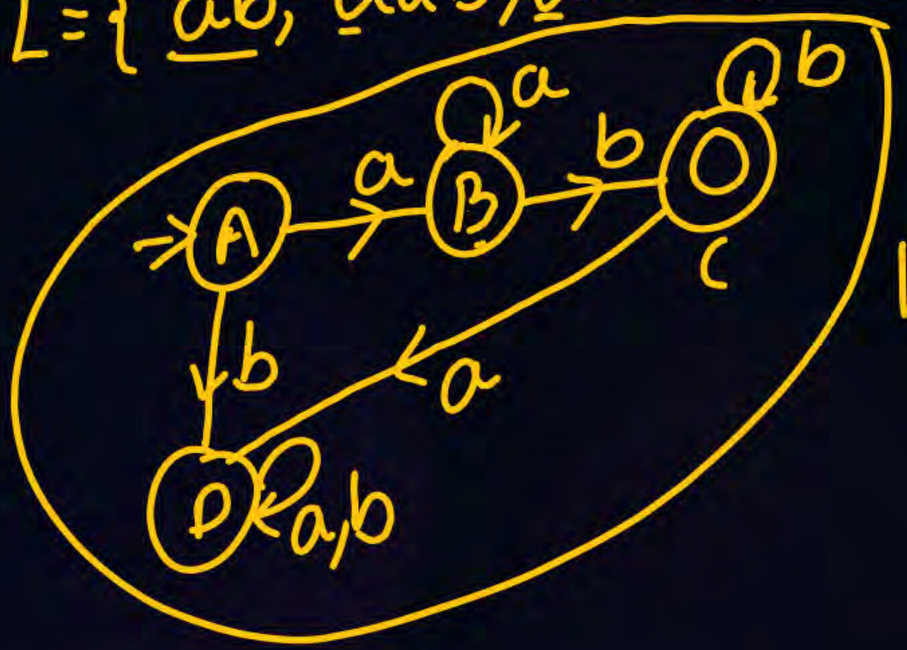
$a^n / n \geq 1 = \{a, aa, aaa \ldots\ldots\}$

$= a^+$

$= aa^*$

$= a\{\epsilon, a, aa, aaa \ldots\ldots\}$

$= \{a, aa, aaa, \ldots\ldots\}$



10 sec

$a^n b^m / n, m \geq 0$

$a^* b^*$

a^* b^*, ϵ

$a^* \subseteq a^* b^*$ ✓

$b^* \subseteq a^* b^*$ ✓

$a^* \cup b^* \neq a^* b^*$ ✓

Rule :- a should not come after b



$a^* b^* \rightarrow a^*, b^*, ab, aab, abb, \cdots$ ✓

a^*

$aa \overset{+}{\cup} bb = \{aa, bb\}$

$aa \cdot bb = aabb$

concat

$a^* \cup b^*$ ab

$a^*b^* \rightarrow a^*, b^*, ab, aab, abb$

Properties of RL

$a^* + b^* \neq a^* b^*$

$\Sigma = \{a, b\}$ $(a+b)^*$ → universal language → Contains every string



$a^* \subseteq (a+b)^*$

$b^* \subseteq (a+b)^*$

$a^* \cup b^* = (a+b)^*$

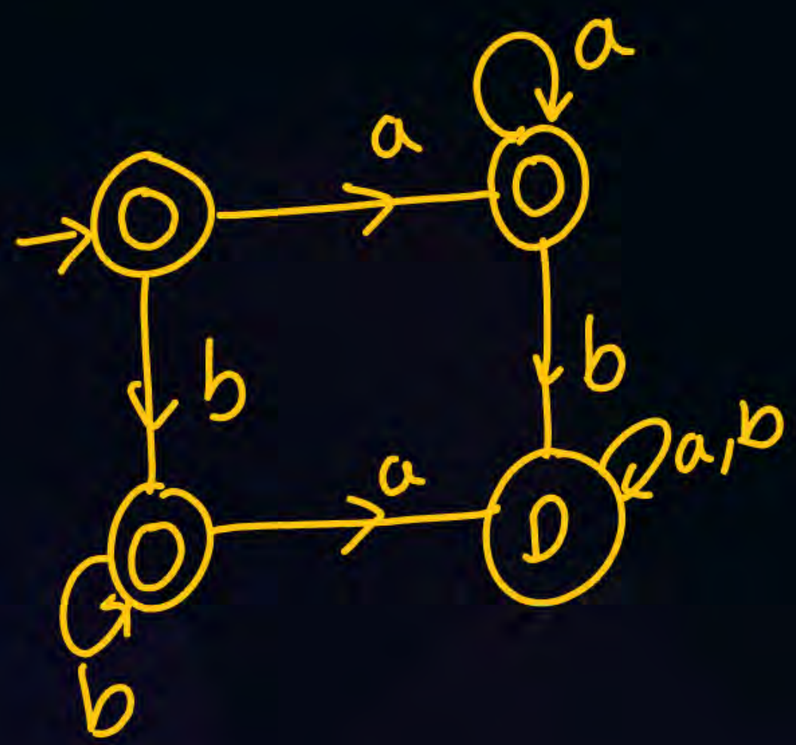
$a^* = \{\epsilon, a, aa, aaa \cdots\}$

$b^* = \{\epsilon, b, bb, bbb \cdots\}$

$ab, ba, aabb$

$a^* + b^*$

No b after 'a' and no 'a' after 'b'



$a^n b^n / n \geq 0$

$\times \times$

$a^{*} b^{*} \neq a^{n} b^{n}$

$a^n b^m / n \geq 0, m \geq 1$

$a^* bb^*$ = b



any no of a's f

$(a+b)^* b$

$a^* b^* b$

$(a+b)^* \neq a^* b^*$

$b^+ = b^* b = bb^*$

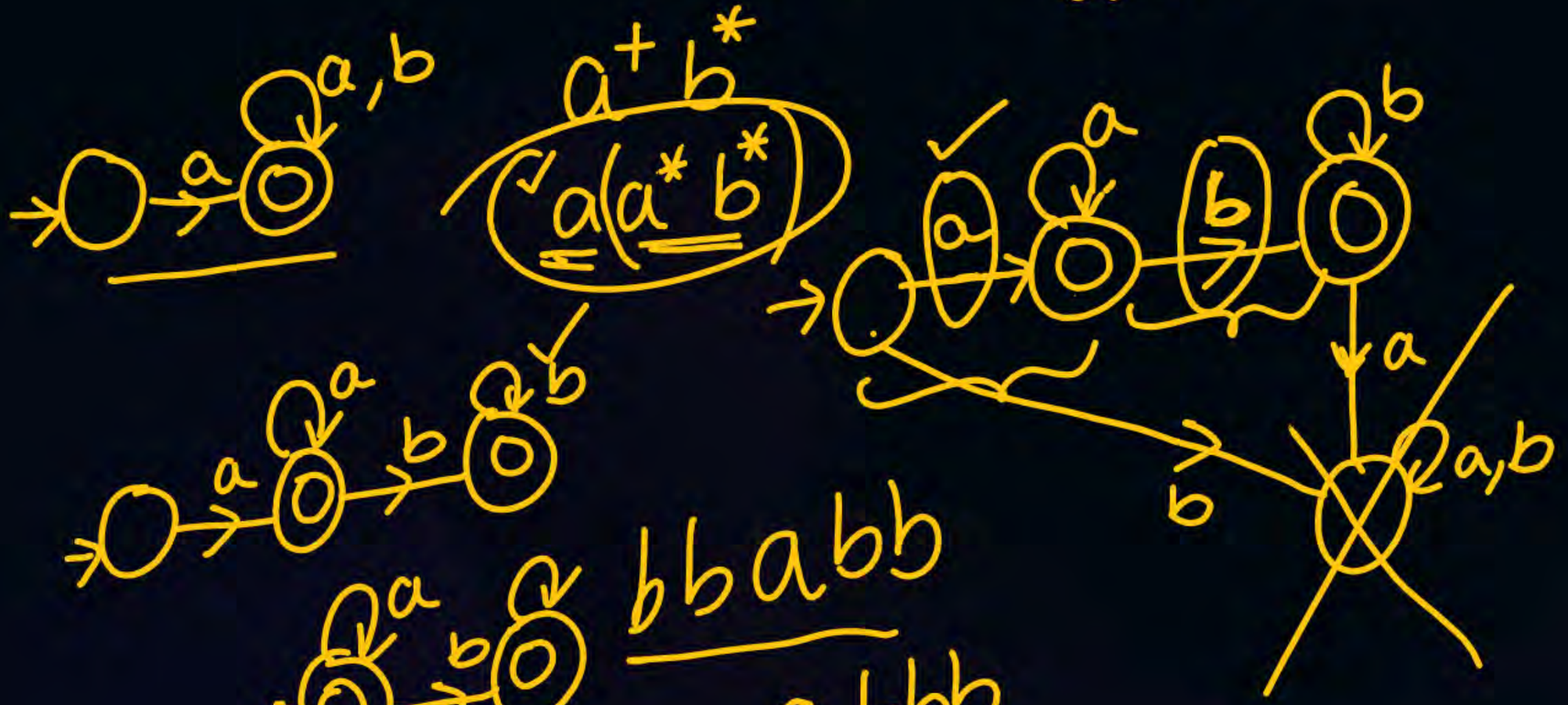
$a^* b^+$

~~any no of a's followed by atleast 1b.~~

$a^n b^m$ / $n \geq 1, m \geq 0$ $\quad a^+ b^*$

$a(a^* b^*)$ $\quad aa^* b^*$

$a^* a b^*$



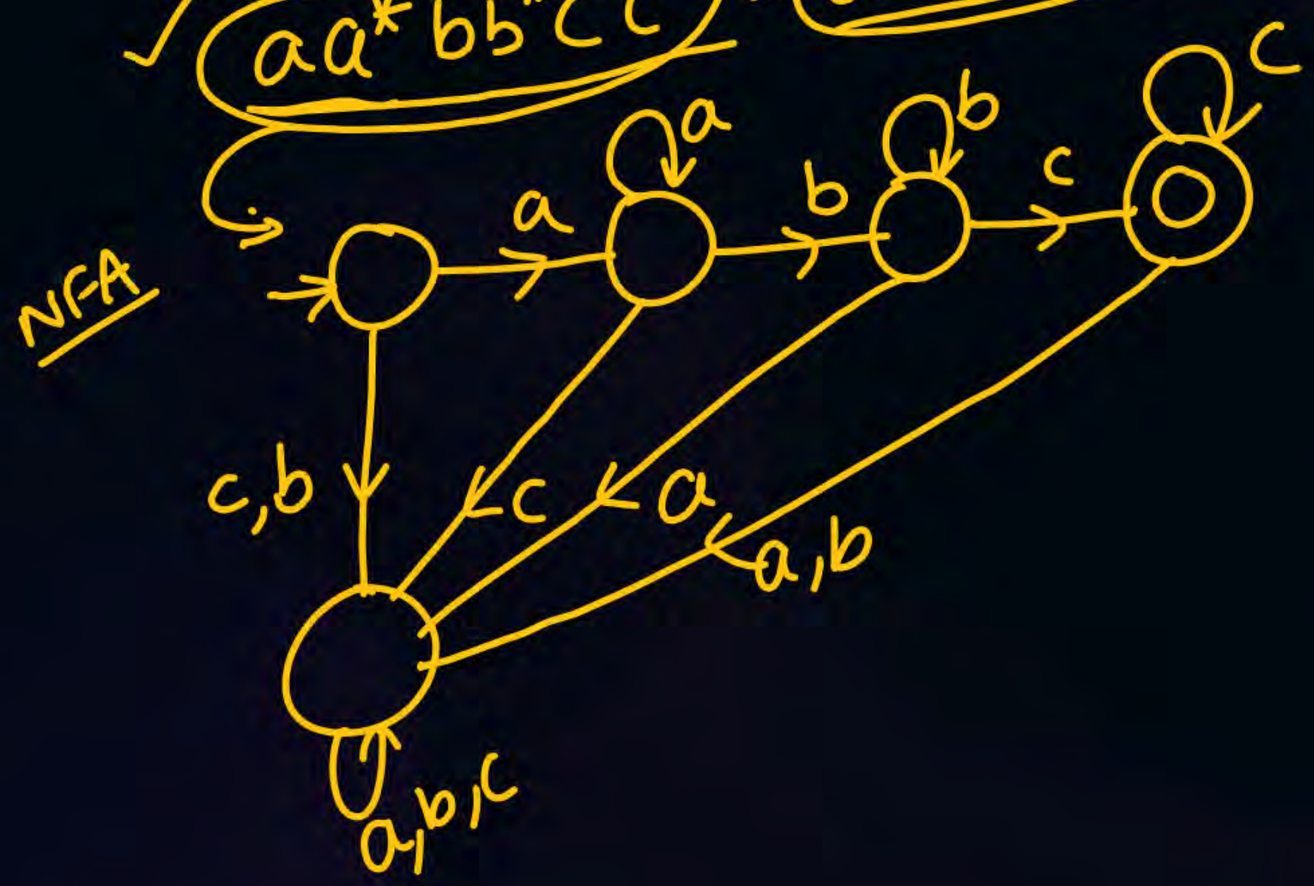
bbabb

abbb

$a^n b^m c^l / n, m, l \geq 1$

$a^+ b^+ c^+$

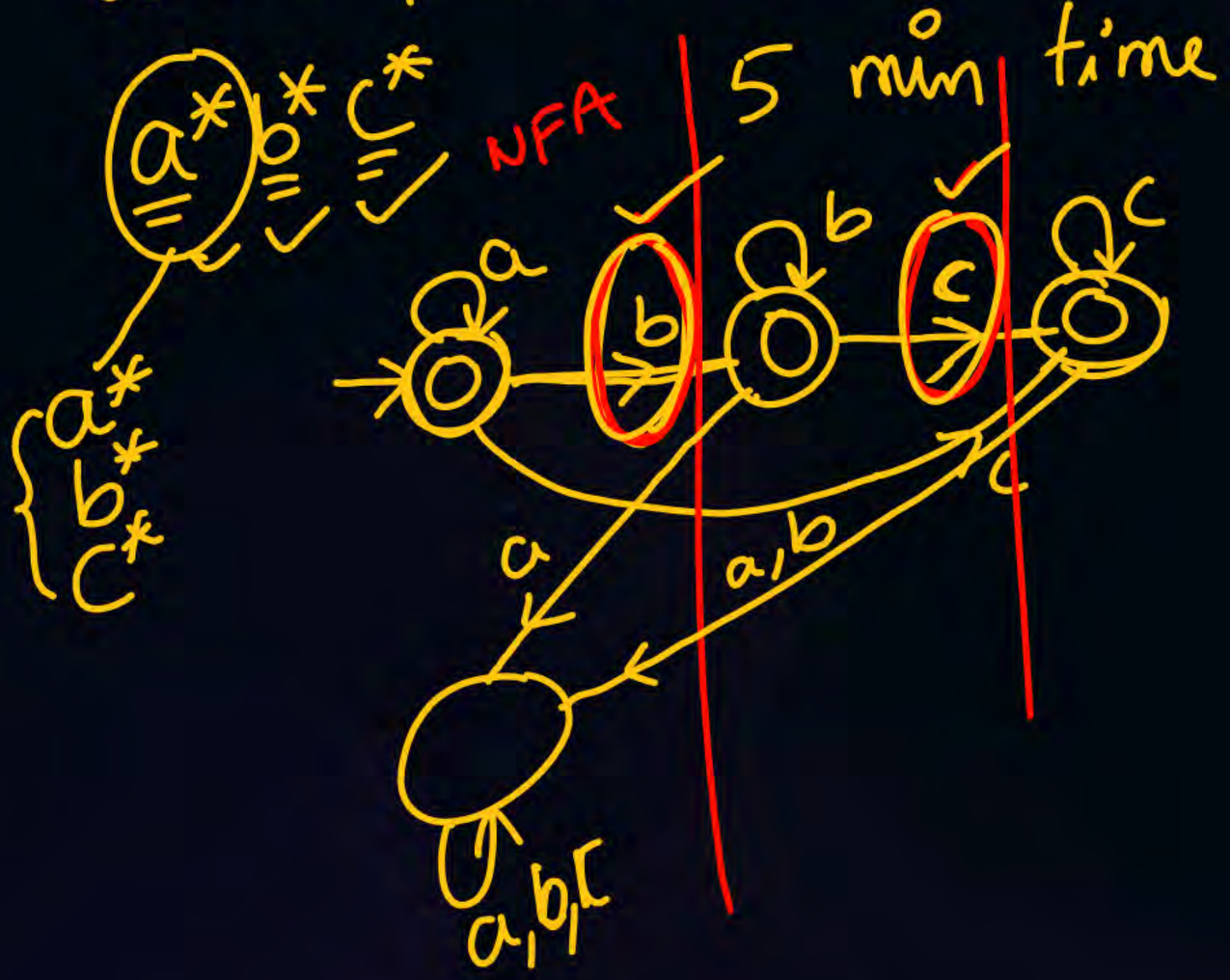
✓ $aa^*bb^*cc^* = a^*ab^*bc^*c$

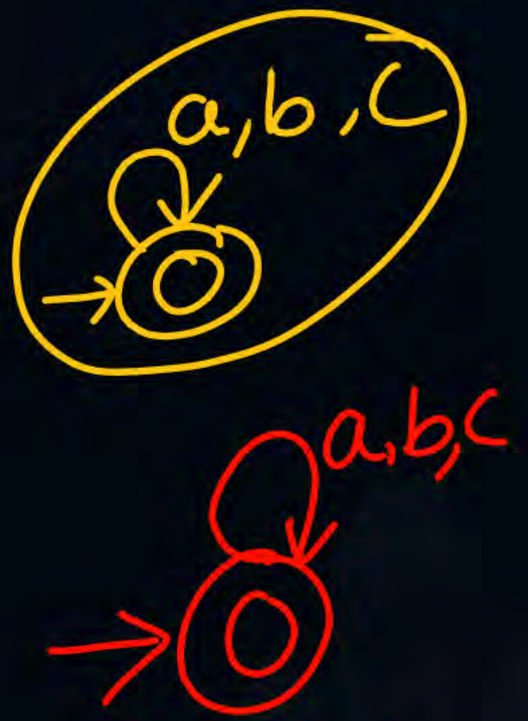


abc

$L = \{abc, aabc, abbc, abbcc, \cdots\}$

$a^n b^m c^l / n, m, l \geq 0$



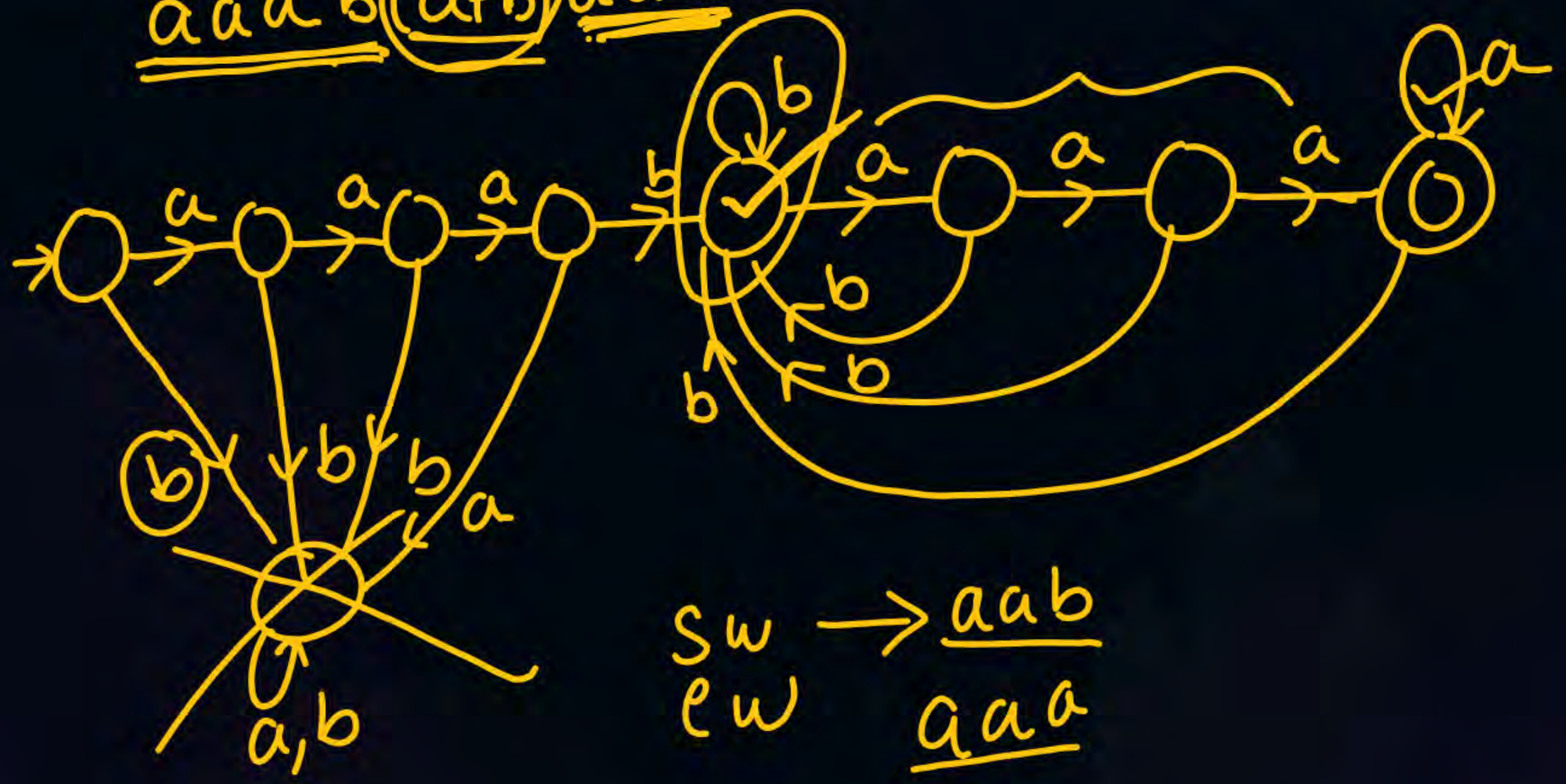


ba

cab

$\Sigma = \{a, b\}$ $L = \{a^3 b (w) a^3 / w \in (a,b)^*\}$

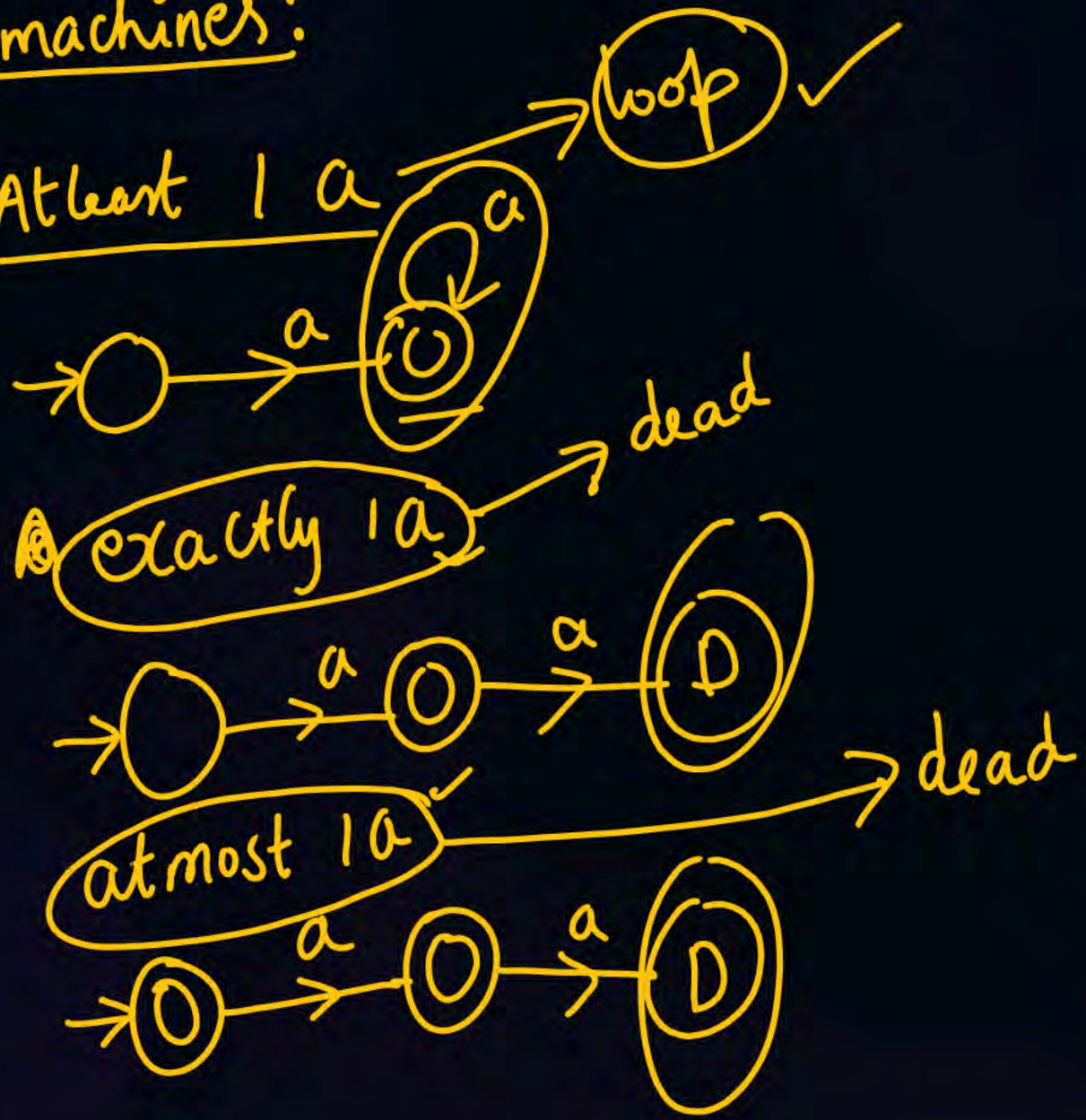
$aaa\,b\,(a+b)^*\,aaa$



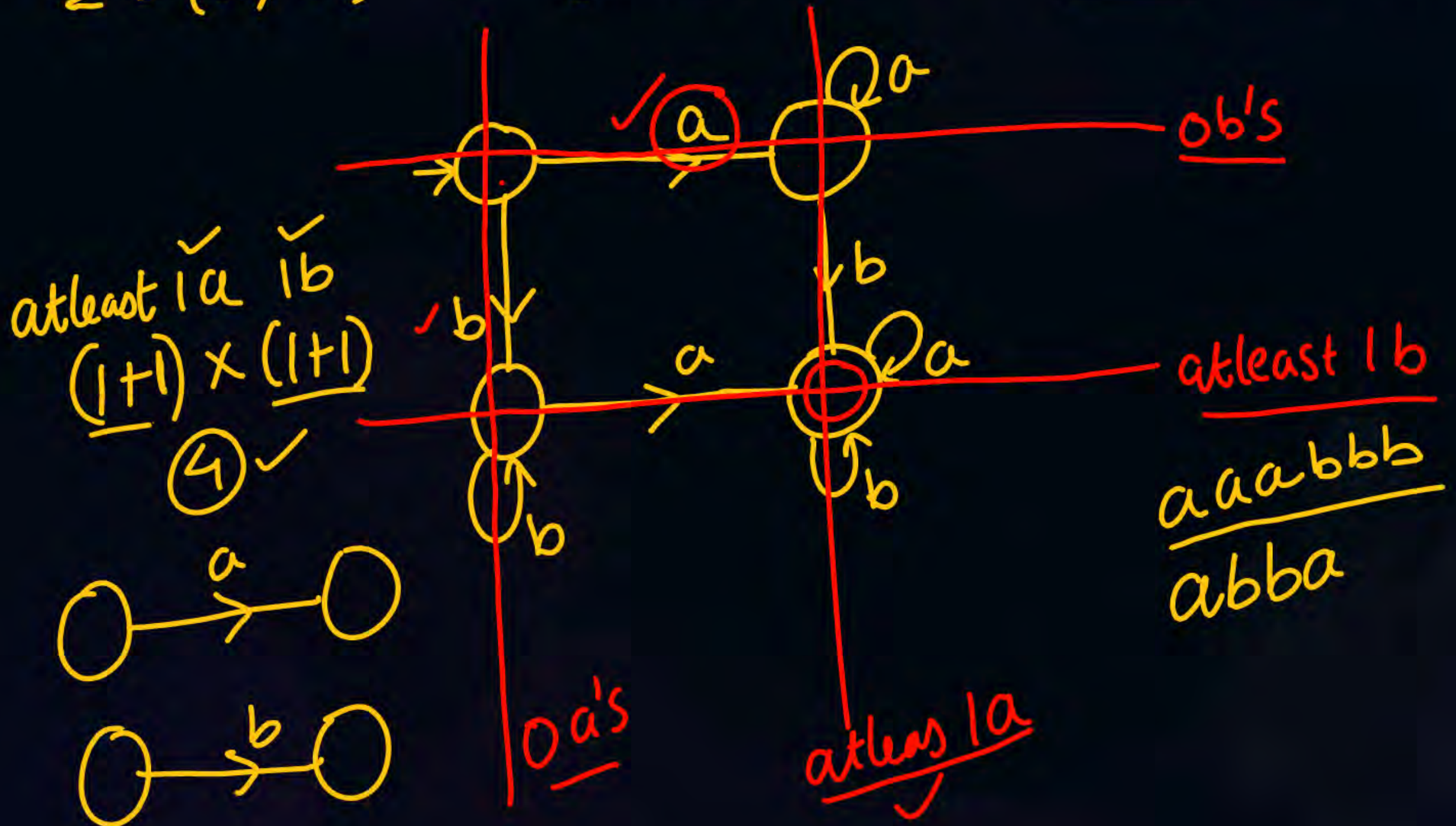
sw → aab

ew → aaa

Grid machines:



$\Sigma = \{a, b\}$ $L = \{w / w$ contains atleast 1a and atleast 1b$\}$



L/L Contains atleast 2 a's and atleast 2 b's

no of states
atleast m a's
atleast n b's



(2+1)



(2+1)

(m+1)(n+1)

3x3 = 9


a
a
a
0b's
b
b
b
a
a
a
1b's
b
b
b
a
a
a
Atleast 2 b's
b
b
b
0 a's
1 a's
atleast 2 a's

THANK - YOU